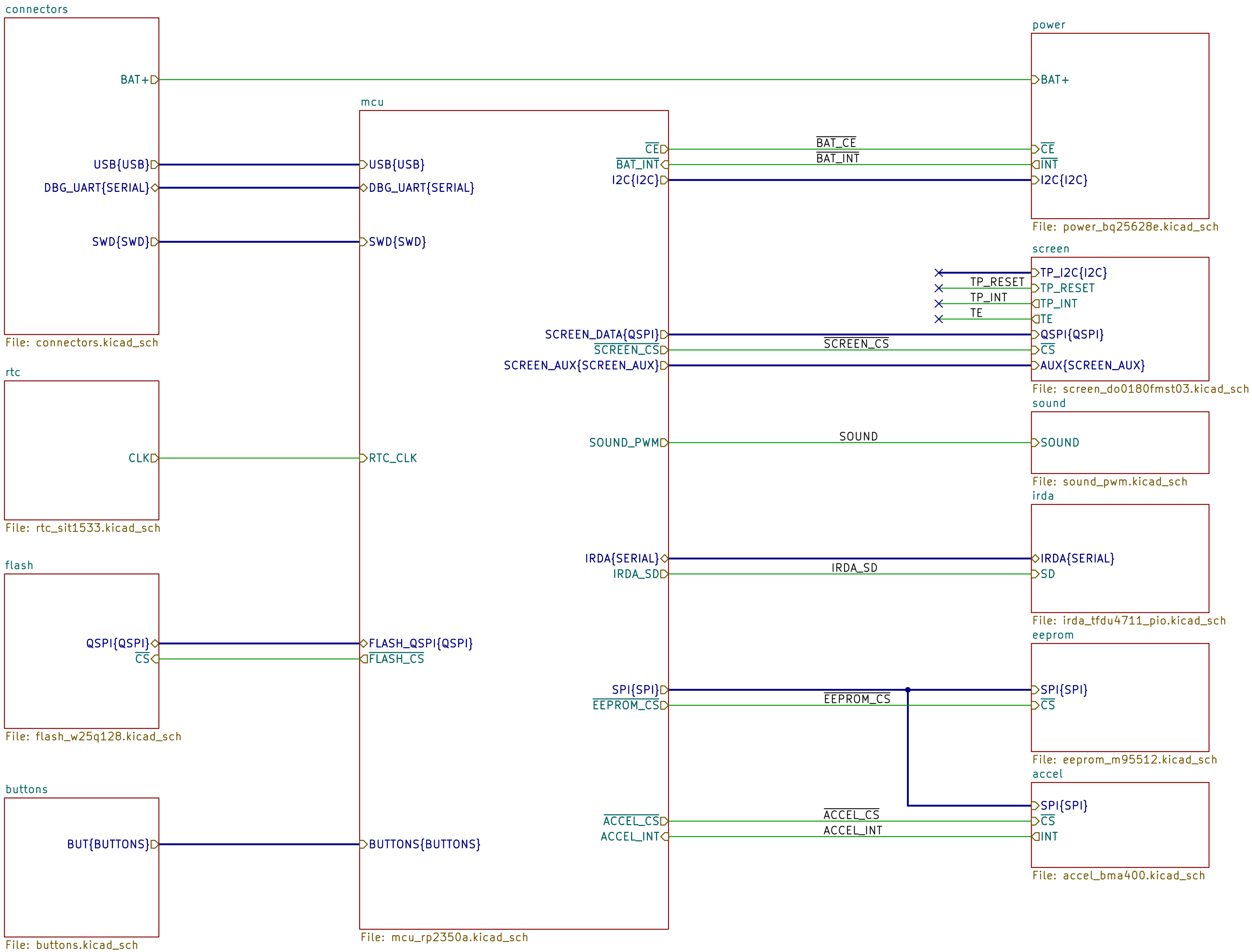
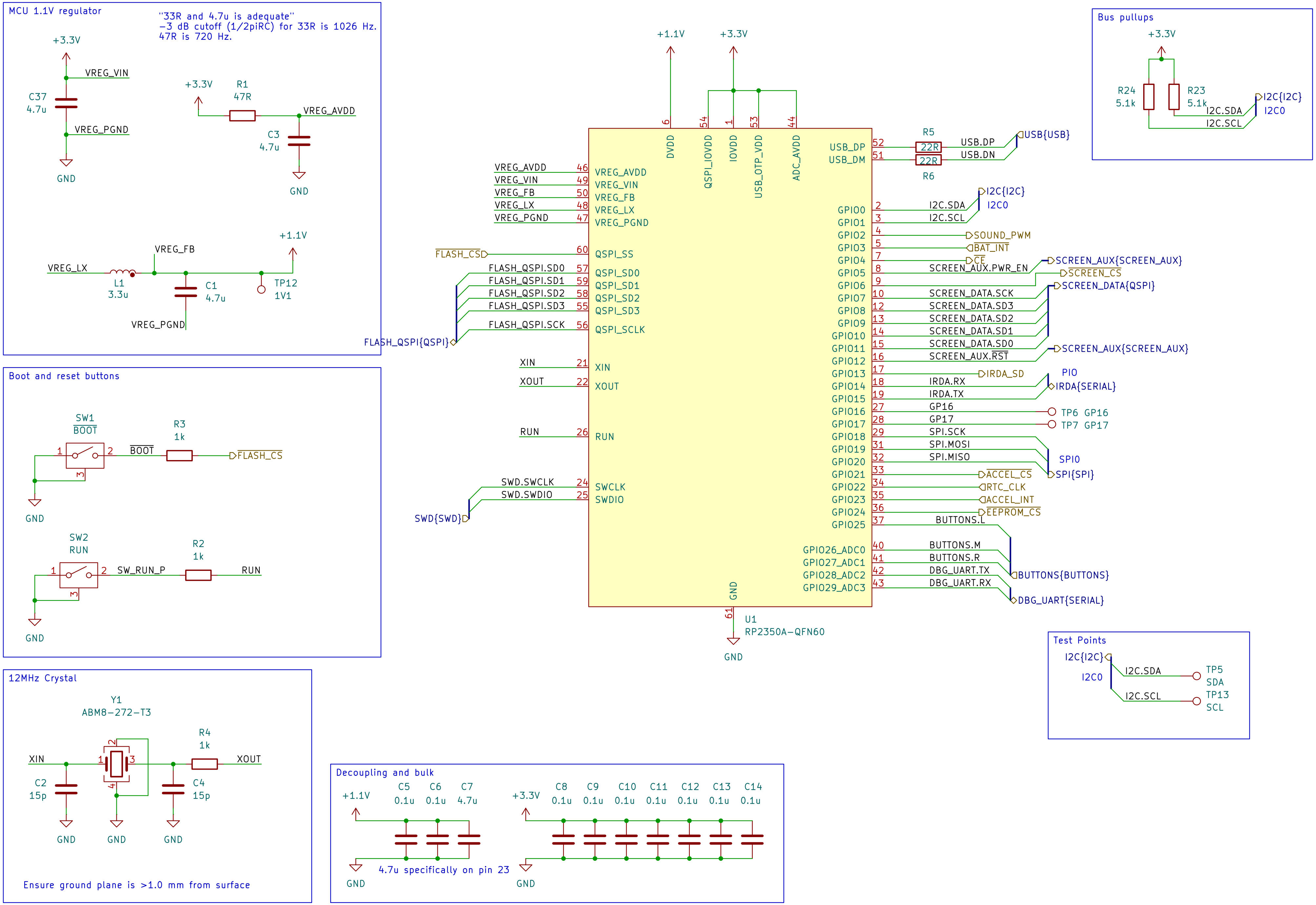
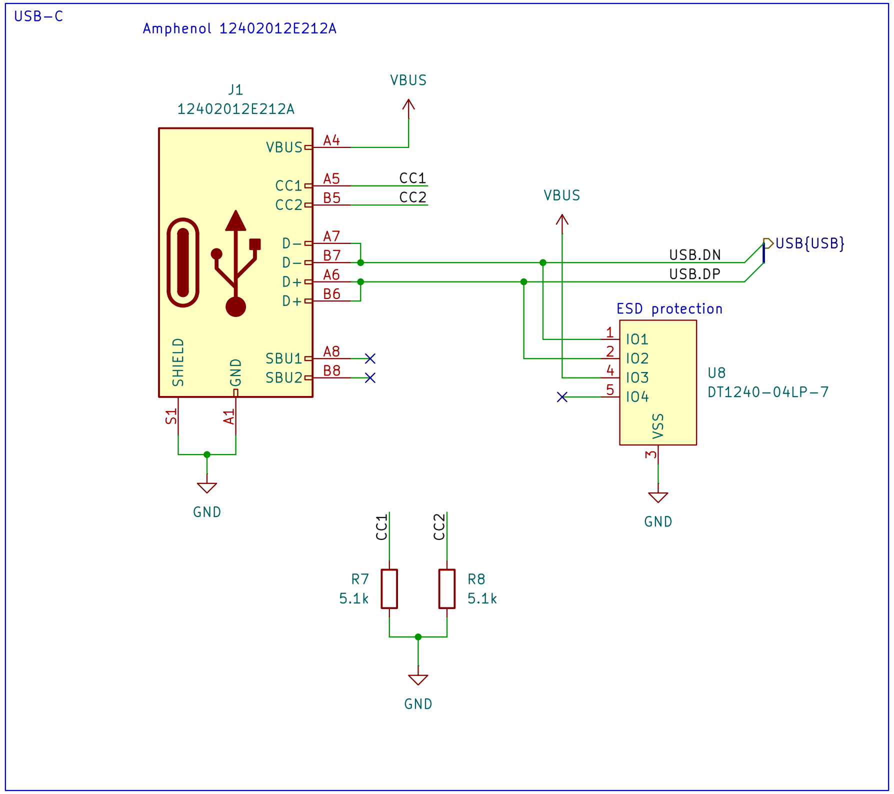
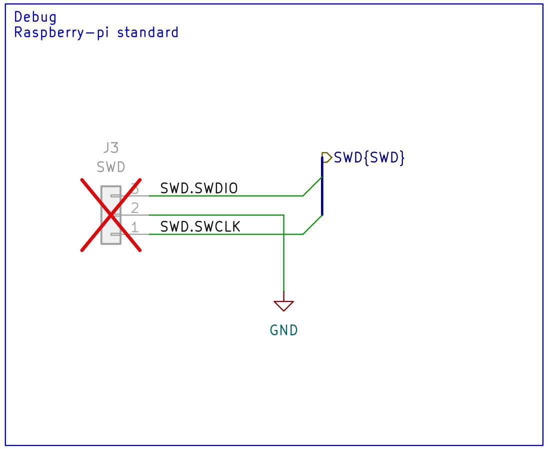
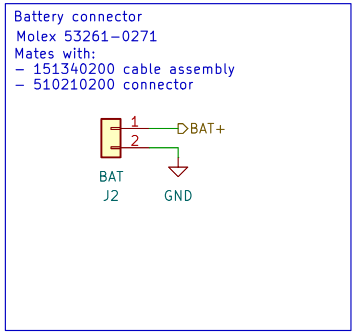
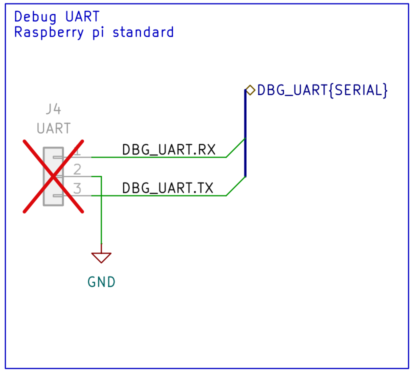
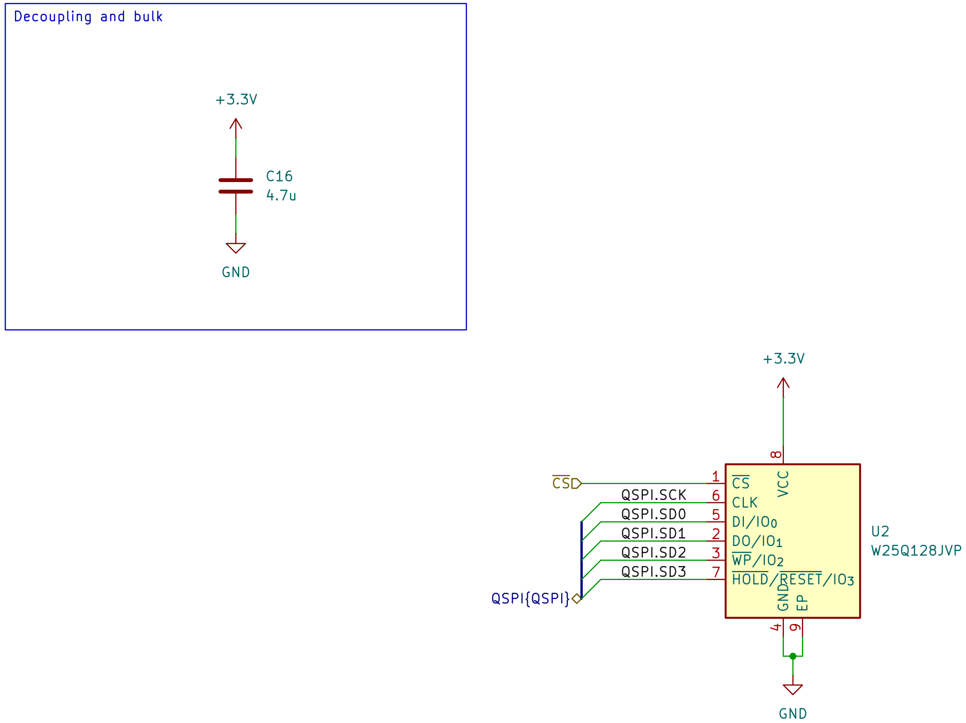
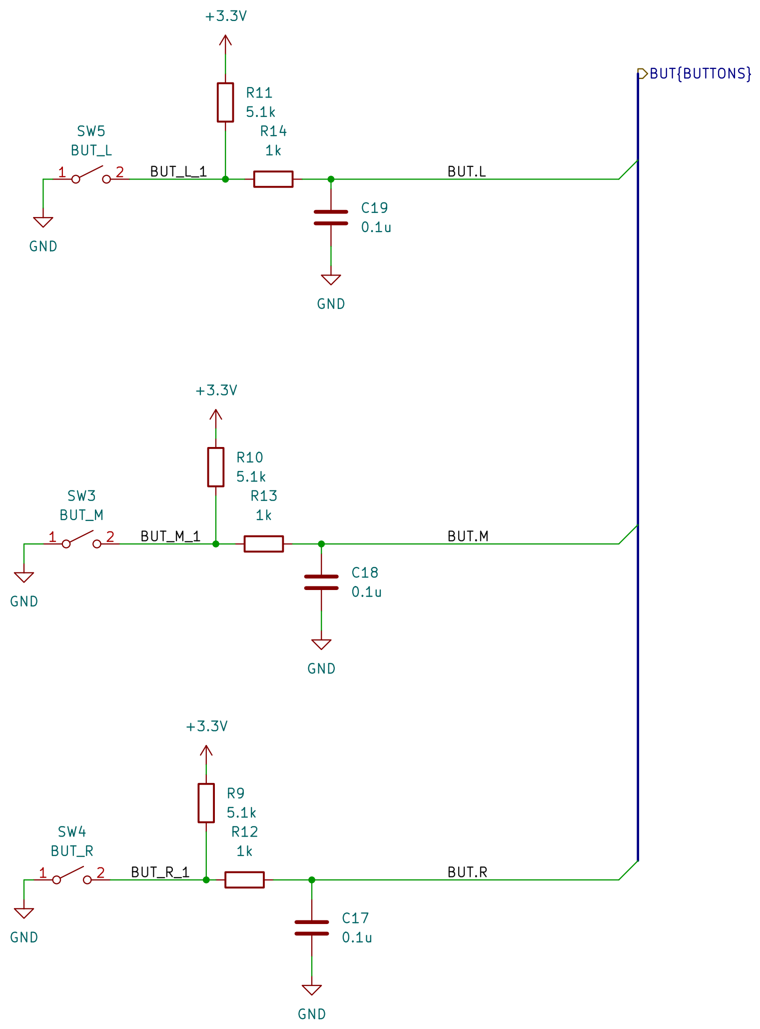

connectors
BAT+
USB{USB}
DBG_UART{SERIAL}
SWD{SWD}
File: connectors.kicad_sch
mcu
USB{USB}
DBG_UART{SERIAL}
SWD{SWD}
RTC_CLK
FLASH_QSPI{QSPI}
FLASH_CS
BUTTONS{BUTTONS}
CE
BAT_INT
I2C{I2C}
SCREEN_DATA{QSPI}
SCREEN_CS
SCREEN_AUX{SCREEN_AUX}
SOUND_PWM
IRDA{SERIAL}
IRDA_SD
SPI{SPI}
EEPROM_CS
ACCEL_CS
ACCEL_INT
File: mcu_rp2350a.kicad_sch
power
BAT+
CE
INT
I2C{I2C}
BAT_CE
BAT_INT
File: power_bq25628e.kicad_sch
screen
TP_I2C{I2C}
TP_RESET
TP_INT
TE
QSPI{QSPI}
CS
AUX{SCREEN_AUX}
SCREEN_CS
File: screen_do0180fmst03.kicad_sch
sound
SOUND
File: sound_pwm.kicad_sch
irda
IRDA{SERIAL}
SD
IRDA_SD
File: irda_tfdu4711_pio.kicad_sch
eeprom
SPI{SPI}
CS
EEPROM_CS
File: eeprom_m95512.kicad_sch
accel
SPI{SPI}
CS
INT
ACCEL_CS
ACCEL_INT
File: accel_bma400.kicad_sch
rtc
CLK
File: rtc_sit1533.kicad_sch
flash
QSPI{QSPI}
CS
File: flash_w25q128.kicad_sch
buttons
BUT{BUTTONS}
File: buttons.kicad_sch



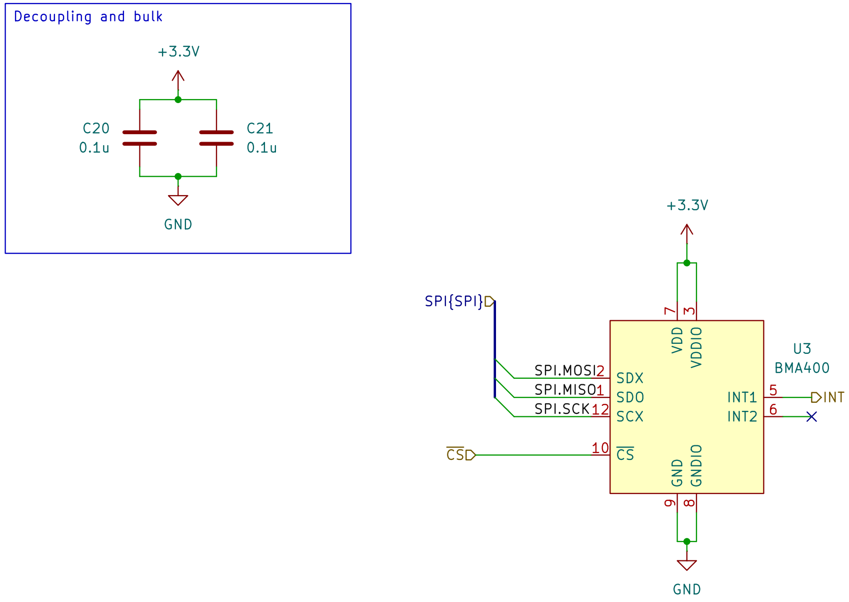
MCU 1.1V regulator
"33R and 4.7u is adequate"
-3 dB cutoff (1/2piRC) for 33R is 1026 Hz.
47R is 720 Hz.
+3.3V
VREG_VIN
C37
4.7u
VREG_PGND
GND
+3.3V
R1
47R
VREG_AVDD
C3
4.7u
GND
VREG_FB
+1.1V
VREG_LX
L1
3.3u
C1
4.7u
TP12
1V1
VREG_PGND
Boot and reset buttons
SW1
BOOT
R3
1k
BOOT
FLASH_CS
GND
SW2
RUN
R2
1k
SW_RUN_P
RUN
GND
12MHz Crystal
Y1
ABM8-272-T3
R4
1k
XIN
XOUT
C2
15p
C4
15p
GND
GND
GND
Ensure ground plane is >1.0 mm from surface
Decoupling and bulk
+1.1V
C5 C6 C7
0.1u 0.1u 4.7u
+3.3V
C8 C9 C10 C11 C12 C13 C14
0.1u 0.1u 0.1u 0.1u 0.1u 0.1u 0.1u
4.7u specifically on pin 23
GND
GND
+1.1V
+3.3V
DVDD
QSPI_IOVDD
IOVDD
USB_OTP_VDD
ADC_AVDD
VREG_AVDD 46 VREG_AVDD
VREG_VIN 49 VREG_VIN
VREG_FB 50 VREG_FB
VREG_LX 48 VREG_LX
VREG_PGND 47 VREG_PGND
FLASH_CS 60 QSPI_SS
FLASH_QSPI.SD0 57 QSPI_SD0
FLASH_QSPI.SD1 59 QSPI_SD1
FLASH_QSPI.SD2 58 QSPI_SD2
FLASH_QSPI.SD3 55 QSPI_SD3
FLASH_QSPI.SCK 56 QSPI_SCLK
FLASH_QSPI{QSPI}
XIN 21 XIN
XOUT 22 XOUT
RUN 26 RUN
SWD.SWCLK 24 SWCLK
SWD.SWDIO 25 SWDIO
SWD{SWD}
GND 61
U1
RP2350A-QFN60
GND
USB_DP 52 R5 22R USB.DP
USB_DM 51 R6 22R USB.DN
USB{USB}
GPIO0 2 I2C.SDA
GPIO1 3 I2C.SCL
I2C{I2C}
I2C0
GPIO2 4 SOUND_PWM
GPIO3 5 BAT_INT
GPIO4 7 CE
GPIO5 8 SCREEN_AUX.PWR_EN
SCREEN_AUX{SCREEN_AUX}
GPIO6 9 SCREEN_CS
GPIO7 10 SCREEN_DATA.SCK
GPIO8 12 SCREEN_DATA.SD3
GPIO9 13 SCREEN_DATA.SD2
GPIO10 14 SCREEN_DATA.SD1
GPIO11 15 SCREEN_DATA.SD0
SCREEN_DATA{QSPI}
GPIO12 16 SCREEN_AUX.RST
SCREEN_AUX{SCREEN_AUX}
GPIO13 17 IRDA_SD
GPIO14 18 IRDA.RX
GPIO15 19 IRDA.TX
PIO
IRDA{SERIAL}
GPIO16 27 GP16 TP6 GP16
GPIO17 28 GP17 TP7 GP17
GPIO18 29 SPI.SCK
GPIO19 31 SPI.MOSI
GPIO20 32 SPI.MISO
SPI0
SPI{SPI}
GPIO21 33 ACCEL_CS
GPIO22 34 RTC_CLK
GPIO23 35 ACCEL_INT
GPIO24 36 EEPROM_CS
GPIO25 37 BUTTONS.L
GPIO26_ADC0 40 BUTTONS.M
GPIO27_ADC1 41 BUTTONS.R
BUTTONS{BUTTONS}
GPIO28_ADC2 42 DBG_UART.TX
GPIO29_ADC3 43 DBG_UART.RX
DBG_UART{SERIAL}
Bus pullups
+3.3V
R24
5.1k
R23
5.1k
I2C.SDA
I2C.SCL
I2C{I2C}
I2C0
Test Points
I2C{I2C}
I2C0
I2C.SDA
TP5
SDA
I2C.SCL
TP13
SCL


USB−C
Amphenol 12402012E212A
J1
12402012E212A
VBUS
VBUS A4
CC1 A5 CC1
CC2 B5 CC2
D− A7
D− B7
D+ A6
D+ B6
USB.DN
USB.DP
USB{USB}
SBU1 A8
SBU2 B8
SHIELD
GND
S1
A1
GND
VBUS
ESD protection
1 IO1
2 IO2
4 IO3
5 IO4
VSS
3
U8
DT1240−04LP−7
GND
CC1
CC2
R7
5.1k
R8
5.1k
GND

Debug
Raspberry−pi standard
J3
SWD
SWD{SWD}
3 SWD.SWDIO
2
1 SWD.SWCLK
GND

Battery connector
Molex 53261−0271
Mates with:
− 151340200 cable assembly
− 510210200 connector
1 BAT+
2
BAT
J2
GND

Debug UART
Raspberry pi standard
DBG_UART{SERIAL}
J4
UART
1 DBG_UART.RX
2
3 DBG_UART.TX
GND

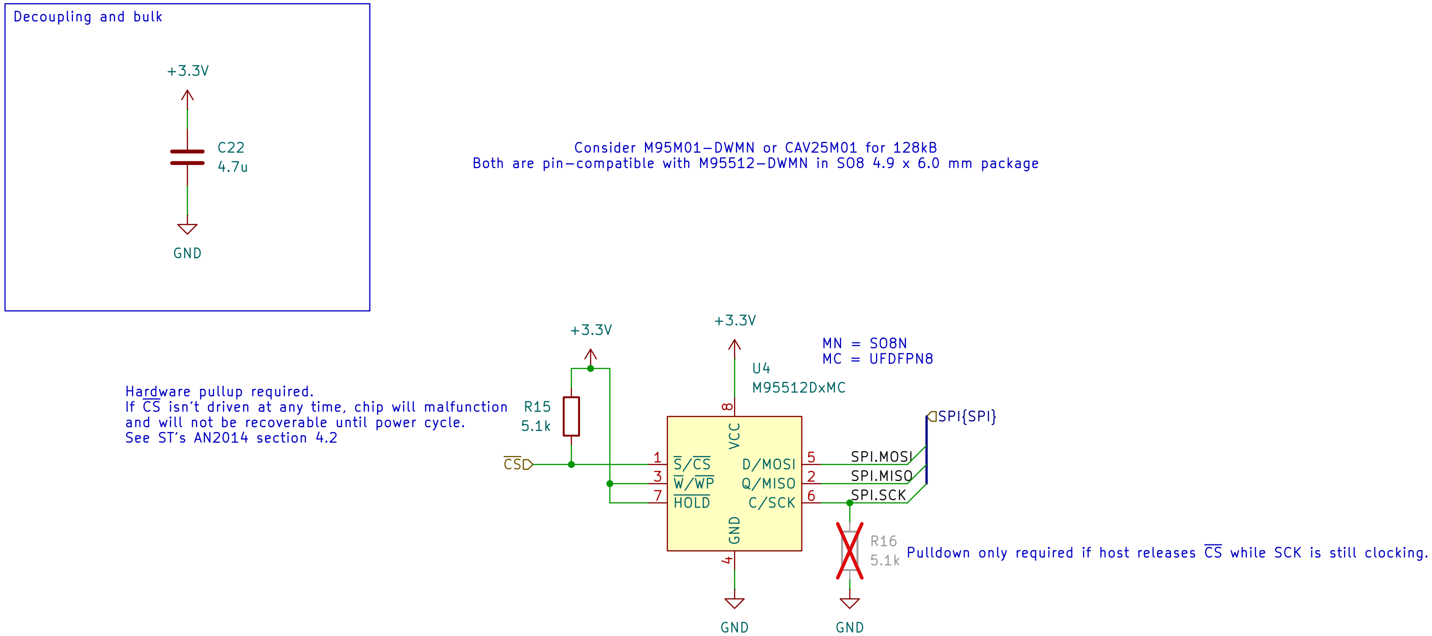

Decoupling and bulk
+3.3V
C16
4.7u
GND
+3.3V
CS
QSPI.SCK
QSPI.SD0
QSPI.SD1
QSPI.SD2
QSPI.SD3
QSPI{QSPI}
1
6
5
2
3
7
8
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
GND
EP
4
9
U2
W25Q128JVP
GND

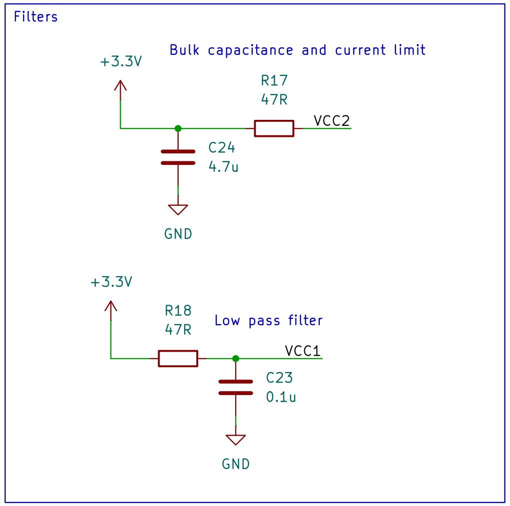


+3.3V
R11
5.1k
R14
1k
SW5
BUT_L
1
2
BUT_L_1
BUT.L
C19
0.1u
GND
GND
BUT{BUTTONS}
+3.3V
R10
5.1k
R13
1k
SW3
BUT_M
1
2
BUT_M_1
BUT.M
C18
0.1u
GND
GND
+3.3V
R9
5.1k
R12
1k
SW4
BUT_R
1
2
BUT_R_1
BUT.R
C17
0.1u
GND
GND



Decoupling and bulk
+3.3V
C20
0.1u
C21
0.1u
GND
+3.3V
SPI{SPI}
SPI.MOSI
SPI.MISO
SPI.SCK
CS
VDD
VDDIO
SDX
SDO
SCX
INT1
INT2
INT
U3
BMA400
GND
GNDIO
GND


Decoupling and bulk
+3.3V
C22
4.7u
GND
Consider M95M01−DWMN or CAV25M01 for 128kB
Both are pin−compatible with M95512−DWMN in SO8 4.9 x 6.0 mm package
+3.3V
+3.3V
MN = SO8N
MC = UFDFPN8
U4
M95512DxMC
Hardware pullup required.
If $\overline{CS}$ isn't driven at any time, chip will malfunction
and will not be recoverable until power cycle.
See ST's AN2014 section 4.2
R15
5.1k
SPI{SPI}
$\overline{CS}$
$\overline{S}/\overline{CS}$
$\overline{W}/\overline{WP}$
$\overline{HOLD}$
VCC
GND
D/MOSI
Q/MISO
C/SCK
SPI.MOSI
SPI.MISO
SPI.SCK
R16
5.1k
Pulldown only required if host releases $\overline{CS}$ while SCK is still clocking.
GND
GND



See Vishay's app note on reference layouts
https://www.vishay.com/docs/82610/referencelayoutscircuitdiagrams.pdf

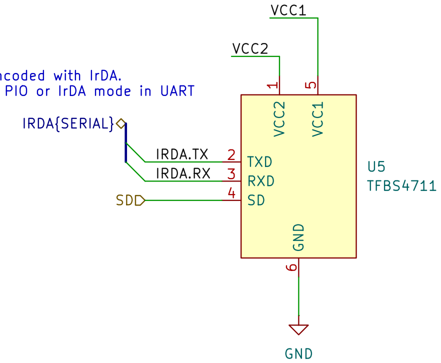
R17 is a current limiting resistor.

47R resistor gives IRED forward current of around 40–50 mA at 3.3 V.
This is 132–150 mW, more than the common 1/16 W (62.5 mW) power rating for resistors.
From testing, this seems to be fine since its pulsed.
Intensity is at around 15%.

100R resistor is 20 mA, or 66 mW, and <10% intensity.

Datasheet says to expect <2 mA average current whtn I_{IRED} (IRED forward current)
is 300 mA and a 20% duty cycle.

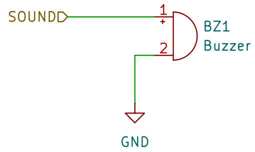
Expect serial to be encoded with IrDA.
MCU can do this with PIO or IrDA mode in UART

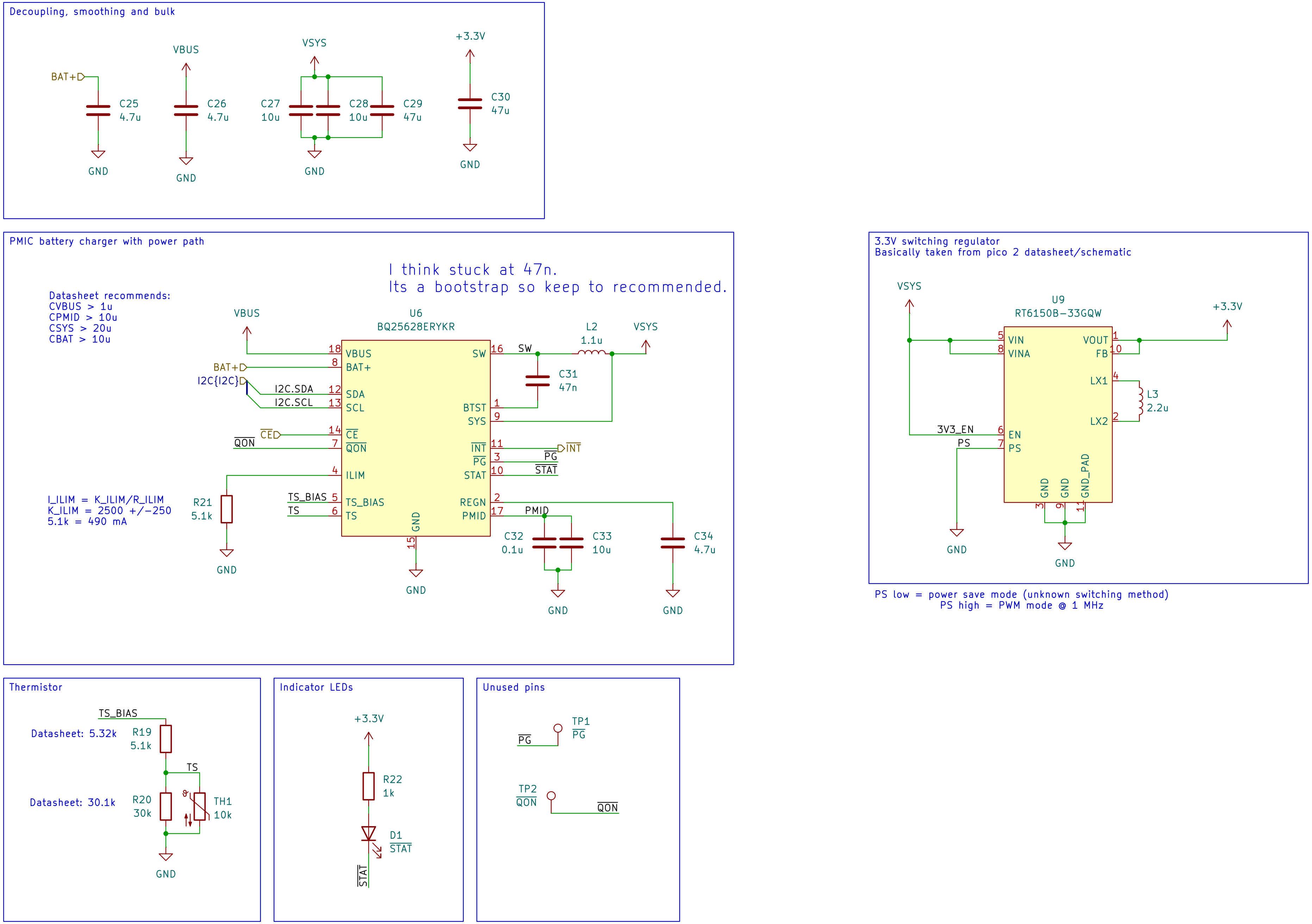


Original uses IrDA SIR V1.0 at 115200 baud, so we do the same.

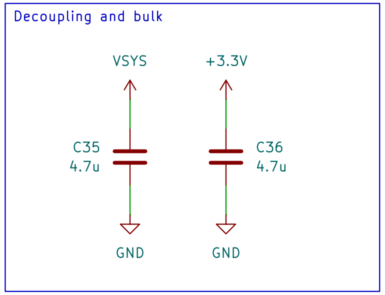
From app note under "emitter signal timing"

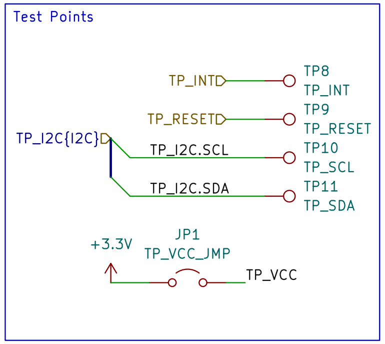
"since the receiver saturates during transmission, it is also necessary to include a latency time of > 150 µs after transmission
before the receiver is ready for reception again"

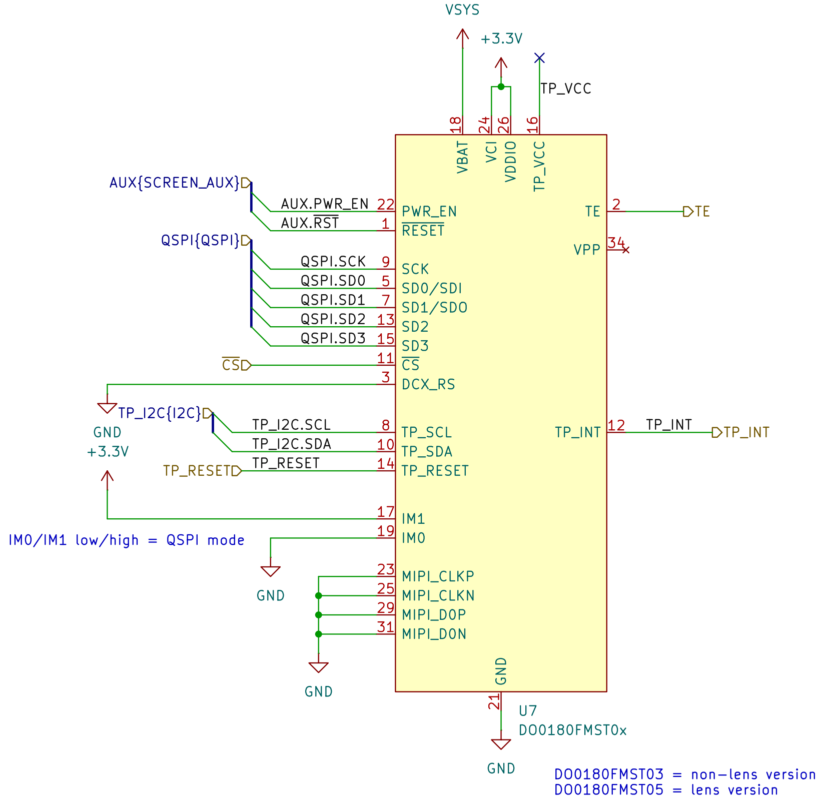

SOUND
1
+
2
BZ1
Buzzer
GND

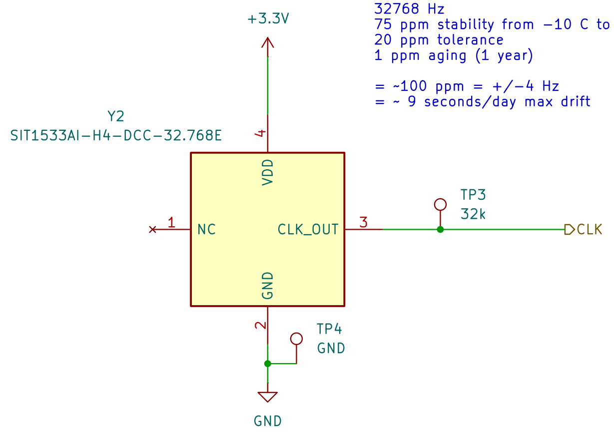


Decoupling, smoothing and bulk
BAT+
C25 4.7u
VBUS
C26 4.7u
VSYS
C27 10u
C28 10u
C29 47u
+3.3V
C30 47u
GND
PMIC battery charger with power path
I think stuck at 47n.
Its a bootstrap so keep to recommended.
Datasheet recommends:
CVBUS > 1u
CPMID > 10u
CSYS > 20u
CBAT > 10u
U6
BQ25628ERYKR
VBUS
BAT+
SDA
SCL
CE
QON
ILIM
TS_BIAS
TS
GND
SW
BTST
SYS
INT
PG
STAT
REGN
PMID
I2C{I2C}
I2C.SDA
I2C.SCL
L2 1.1u
C31 47n
C32 0.1u
C33 10u
C34 4.7u
I_ILIM = K_ILIM/R_ILIM
K_ILIM = 2500 +/-250
5.1k = 490 mA
R21 5.1k
3.3V switching regulator
Basically taken from pico 2 datasheet/schematic
U9
RT6150B-33GQW
VIN
VINA
EN
PS
VOUT
FB
LX1
LX2
GND_PAD
3V3_EN
L3 2.2u
PS low = power save mode (unknown switching method)
PS high = PWM mode @ 1 MHz
Thermistor
TS_BIAS
Datasheet: 5.32k
R19 5.1k
Datasheet: 30.1k
R20 30k
TH1 10k
Indicator LEDs
R22 1k
D1 STAT
Unused pins
TP1 PG
TP2 QON


Decoupling and bulk
VSYS
+3.3V
C35
4.7u
C36
4.7u
GND
GND


Test Points
TP_INT
TP8
TP_INT
TP_RESET
TP9
TP_RESET
TP_I2C{I2C}
TP_I2C.SCL
TP10
TP_SCL
TP_I2C.SDA
TP11
TP_SDA
+3.3V
JP1
TP_VCC_JMP
TP_VCC


VSYS
+3.3V
TP_VCC
18 VBAT
24 VCI
26 VDDIO
16 TP_VCC
AUX{SCREEN_AUX}
AUX.PWR_EN 22 PWR_EN
AUX.RST 1 RESET
QSPI{QSPI}
QSPI.SCK 9 SCK
QSPI.SD0 5 SD0/SDI
QSPI.SD1 7 SD1/SDO
QSPI.SD2 13 SD2
QSPI.SD3 15 SD3
CS 11 CS
3 DCX_RS
GND
TP_I2C{I2C}
TP_I2C.SCL 8 TP_SCL
TP_I2C.SDA 10 TP_SDA
+3.3V
TP_RESET
TP_RESET 14 TP_RESET
17 IM1
19 IM0
IM0/IM1 low/high = QSPI mode
GND
23 MIPI_CLKP
25 MIPI_CLKN
29 MIPI_D0P
31 MIPI_D0N
GND
TE 2 TE
VPP 34
TP_INT 12 TP_INT TP_INT
GND 21
U7
DO0180FMST0x
GND
DO0180FMST03 = non-lens version
DO0180FMST05 = lens version



32768 Hz
75 ppm stability from -10 C to +70 C
20 ppm tolerance
1 ppm aging (1 year)

= ~100 ppm = +/-4 Hz
= ~ 9 seconds/day max drift